

Tanvi Garg

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EDUCATION

Virginia Tech

M.S. Thesis in Computer Engineering | GPA: 4.0 / 4.0

Honors: CCI Southwest Virginia (SWVA) Scholarship Recipient – 2025, 2026

Arlington, VA

Aug 2023 – Dec 2026

Virginia Tech

M.Eng. in Computer Science | GPA: 3.80 / 4.0

Blacksburg, VA

Aug 2021 – May 2023

NMIMS

B.Tech. in Computer Science and Engineering | GPA: 3.45 / 4.0

Mumbai, India

Jul 2017 – May 2021

RESEARCH EXPERIENCE

Graduate Research Assistant

Networks and Systems Security Lab, Virginia Tech | Advisor: Dr. Angelos Stavrou

Aug 2023 – Present

Arlington, VA

- Identified a novel class of privacy violation in multi-agent LLM pipelines where agents receive out-of-scope information through unfiltered inter-agent message passing; developed a benchmark to measure its prevalence across pipeline configurations.
- Built a middleware enforcement layer that intercepts inter-agent messages and applies role-scoped permission filtering grounded in HIPAA's minimum necessary standard, preventing information exposure without degrading task performance.
- Conducted root-cause and statistical analysis of detection failures in LLM-based prompt injection systems, identifying positional brittleness as a previously undocumented failure mode.
- Analyzed real-time signal-strength telemetry from a 5G NR testbed (OpenAirInterface, USRP X310) for anomaly detection and closed-loop RAN performance optimization.

Research Programmer

Virginia Tech

Jun 2022 – Aug 2023

Arlington, VA

- Developed a Dockerized deployment of Wave, a decentralized authorization framework with transitive delegation, enabling reproducible self-service research environments and reducing onboarding time.
- Emulated a 5G core and gNBSIM RAN in a local Kubernetes cluster (Docker Compose, Helm, Multus CNI, RKE) to study network slices and UE performance; further emulated OpenAirInterface with FlexRIC to explore near-RT RIC and xApp-driven optimization.
- Built structured data pipelines in SQL and Python to ingest real-time telemetry and identify root causes of performance degradation.

INDUSTRY EXPERIENCE

Web Development Intern

Make-A-Wish India

Jul 2020 – Sep 2020

Mumbai, India

- Aligned website feature development with organizational and product goals by collaborating with cross-functional teams, ensuring delivered features met specifications and quality standards.
- Used Google Ads and Analytics to inform iterative, data-driven improvements to user experience, supporting seamless interactions for volunteers, beneficiaries, and staff.

UX Design Intern

Net Solutions

Dec 2018

Chandigarh, India

- Contributed to UX design of a web application, partnering with the product team to translate user feedback into iterative interface improvements.

TEACHING EXPERIENCE

Graduate Teaching Assistant

Aug 2023 – Dec 2023 & Aug 2026 – Present

Network and Protocols, Virginia Tech | TA to: Dr. Nektaria Tryfona

Arlington, VA

- Assisted in delivering course content on TCP/IP, network security protocols, and packet analysis; held office hours and supported graduate students and industry professionals enrolled in the MIT program.

Graduate Teaching Assistant

Jan 2024 – May 2024

Network Application Design, Virginia Tech | TA to: Dr. Carl Harris

Arlington, VA

- Supported instruction on distributed application design and client-server architectures, graded assignments; held office hours and provided technical feedback to graduate students.

PUBLICATIONS

- [1] Garg, T., et al. “A novel approach of privacy-preserving data sharing system through data-tagging with role-based access control.” *World Journal of Engineering* (2021).
- [2] Garg, T. & Kagalwalla, N. “Challenges of Implementing Privacy Policies Across the Globe.” In *Data Protection and Privacy in Healthcare*, pp. 212–228. CRC Press, 2021.
- [3] Garg, T., et al. “A survey on security and privacy issues in IoT.” *International Journal of Electrical & Computer Engineering*, 10(5), 2020.
- [4] Kagalwalla, N., Garg, T., et al. “A survey on implementing Privacy in Healthcare: An Indian perspective.” *International Journal of Advanced Trends in Computer Science and Engineering*, 8(3), 963–982, 2019.

SELECTED PROJECTS

RoleGuard: Preventing Privacy Leakage in Multi-Agent LLM Systems

2026

- Formalized role-boundary leakage, a novel privacy violation in multi-agent LLM pipelines where agents receive PHI beyond their HIPAA-permitted scope.
- Designed RoleGuard, an inference-time middleware enforcing role-scoped access control at every inter-agent boundary without retraining underlying LLM agents.
- Built RoleLeak, a two-tier leakage benchmark, reducing PHI boundary violations from 100% to 0% across 180 mCODE synthetic patients while preserving 100% task completion.
- Evaluated adversarial robustness against prompt injection, role impersonation, and targeted PHI fishing using a cyclic LangGraph pipeline; all attacks blocked by RoleGuard.

O-RAN xApp Development

2023

- Deployed a Near-RT RIC environment (srsRAN, FlexRIC, ZeroMQ) and implemented custom xApps to subscribe to real-time signal-strength metrics via the ORAN-E2SM-KPM service model.
- Diagnosed a persistent UE-to-gNB attachment failure by analyzing message flows in Wireshark, isolating the fault to the ZeroMQ RF driver layer.

OAI 5G NR Standalone Setup with COTS UEs

2023

- Established a 5G NR Standalone environment using OpenAirInterface, integrating and testing connectivity with commercial off-the-shelf devices (OnePlus, Google Pixel).
- Configured USRP X310 radios with high-speed servers to support 10G throughput, including SIM card provisioning for controlled connectivity testing.

TECHNICAL SKILLS

Languages: Python, SQL, R, JavaScript, Bash

ML / NLP / Agentic Systems: PyTorch, Hugging Face Transformers, DeBERTa, Scikit-learn, LangGraph, Llama3 (Ollama), adversarial robustness evaluation, prompt injection analysis, multi-agent pipeline design

Security & Privacy: LINDDUN, STRIDE, threat modeling, HIPAA compliance, role-based access control, mCODE FHIR R4, OWASP, Nmap, Wireshark

Networking: OpenAirInterface (5G Core + RAN), srsRAN, FlexRIC, USRP X310, ZeroMQ, 3GPP standards

Infrastructure: Docker, Kubernetes, Helm, RKE, Multus CNI, AWS, Git, LaTeX, Databricks

Analytics: Statistical Modeling, Root-Cause Analysis, PowerBI